Designing Robots 

Cadette Robotics Badge — Girls Build Bots

Presented by your robotics mentors!

Robots that change lives 💗

Assistive robotics — robots designed for human needs:

- 🦾 Prosthetic limbs & exoskeletons that restore movement
- 👵 Companion robots that reduce isolation for older adults
- 🏥 Hospital robots that deliver medication
- 🚨 Search-and-rescue robots that go where humans can’t
- 🌾 Agricultural robots that feed the world

The engineer’s three questions 🤔

For every assistive robot, ask:

1. **Whose** problem does it solve? *(the user comes first)*
2. What **sensors and actions** does it need?
3. **What could go wrong?** *(safety and failure modes)*

That third question separates beginners from real engineers.

Your Design Challenge 🎯

Teams of 2–3. Choose your mission:

- 🦵 **Mobility assist** — retrieve dropped objects for someone who can't bend down
- 💬 **Companion** — reduce isolation for a homebound older adult
- 💊 **Home care** — medication reminders and assistance
- 🚨 **Rescue** — locate a person where humans can't safely go
- 🖐️ **Accessibility** — open doors/containers for limited hand mobility

Phase 1: Know your user (15 min) 

A mentor will role-play your **user**. Interview them:

- "Walk me through your day. When does this problem show up?"
- "What have you already tried?"
- "What would make a solution annoying instead of helpful?"

Write a **problem statement**: *[User] needs a way to [goal] because [barrier].*

Great engineers design WITH people, not just FOR them.

Phase 2: Two concepts + decision matrix (25 min)

Sketch **TWO different concepts** — committing to your first idea is a rookie mistake.

Then score both in a decision matrix:

Criteria (pick 3)	Concept A	Concept B
Safety	/5	/5
Cost	/5	/5
Ease of use	/5	/5

Highest score wins. Document WHY.

Phase 3: Prototype (40 min) 🔨

Build a physical model representing: - **Sensors** — how it perceives the user & environment - **Controller** — where decisions happen - **Actuation** — how it moves - **The tool** — what actually touches the user's world

Bottle cap = camera. Straw = hydraulic arm. Represent, don't perfect.

Phase 4: Failure-mode analysis (10 min)

The question professionals ask: **“What breaks first?”**

- Battery dies mid-task?
- Sensor gets blocked or dirty?
- User falls or drops it?
- It misreads the situation?

Pick the scariest failure → make **one documented design change** to address it.

Design Review 🎓

Present to the mentor panel. Expect real questions:

> *“Why this sensor and not a cheaper one?” “What happens if the battery dies mid-task?” “How would you make this affordable?”*

“We don’t know yet — here’s how we’d find out” is a legitimate engineering answer. Use it.

What you accomplished ✅

- Interviewed a user and wrote a problem statement
- Compared concepts with a decision matrix
- Built a prototype with all major subsystems
- Ran failure-mode analysis and iterated
- Defended your design under questioning

This is the actual workflow of professional design engineering.

Level up 🚀

- **Tinkercad** (tinkercad.com) — free CAD; design your robot in 3D
- **Instructables** — real build guides with electronics
- **e-NABLE** (enablingthefuture.org) — volunteers 3D-print prosthetic hands
- **EngineerGirl** (engineergirl.org) — profiles of women engineers + contests
- Ask us about CAD on our FRC team!